

# A. S. M. SAMIUL ISLAM

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## RESEARCH INTERESTS

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Human-Computer Interaction (HCI) • User Experience Research • Assistive Technology • Embedded Machine Learning • IoT and Wearable Sensing • Deep Learning • User-Centered Design

## EDUCATION

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**Master of Science in Management of Technology (MOT)** October 2024 – Present

Institute of Appropriate Technology (IAT),

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

- CGPA: 3.67/4.00 | Thesis defence pending
- Thesis: IoT and LLM-based Bangla Sign Language (BdSL) recognition using a wearable smart glove (proposal CASR-approved)
- Supervisor: Tahsina Farah Sanam, Ph.D.

**Bachelor of Science in Computer Science and Engineering** 2019 – 2023

North South University, Dhaka, Bangladesh

- CGPA: 3.59/4.00 | 50% Merit Scholarship | Distinction: **CUM LAUDE**

**Higher Secondary Certificate (HSC), Science** July 2016 – June 2018

Govt. Azizul Haque College, Bogura, Bangladesh | GPA: 5.00/5.00

**Secondary School Certificate (SSC), Science** February 2011 – June 2016

Bogra Zilla School, Bogura, Bangladesh | GPA: 5.00/5.00

## PROFESSIONAL EXPERIENCE

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**Postgraduate Research Fellow** November 2024 – Present

Institute of Appropriate Technology, BUET, Dhaka, Bangladesh

- Developing an IoT and LLM-based Bangla Sign Language recognition system on a wearable sensor glove
- Building ESP32-based sensing hardware with multiple MPU-6050 gyroscopes on a TCA9548A I2C multiplexer
- Building a gesture vocalizer using NLP and ML for real-time gesture-to-speech/text translation
- Member of the ScabieSense innovation project team (scabies detection, treatment guidance, outbreak mapping), now used by doctors screening madrasa students

**UI/UX Designer and Researcher** August 2023 – April 2024

Seed of Intelligence, Mirpur DOHS, Dhaka (Korea-operated)

- Designed AI-powered CCTV-based Video Software as a Service (VSaaS) platform
- Created user flows, schema diagrams, and game UI integrated with VSaaS system
- Designed user diagrams for connected online chat software

**Teaching Assistant, Department of Mathematics and Physics** June 2021 – January 2022

North South University, Dhaka, Bangladesh

- Assisted five professors in course delivery and student assessment
- Provided academic consultation to undergraduate students

## ACHIEVEMENTS

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- **Best Designer Award**, Shaheed Smriti Hall, BUET (August 2026)
- **Incubator Selection**, ICSETEP Project, Department of CSE, BUET (August 2025)
- **UIHP Pre-seed Grant**, Team Arora, Cohort IC5, University Innovation Hub Program at BUET (2024)
- **CUM LAUDE Distinction**, North South University (2022)
- **50% Merit Scholarship**, North South University (2019–2022)
- **17th Position**, Regional 3rd Bangladesh Physics Olympiad
- **Divisional 2nd Runner Up**, 9th and 10th Bangladesh Math Olympiad
- **First Position (Team)**, Acting Competition, BTV Subarna Jayanti

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## RESEARCH EXPERIENCE

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### IoT and LLM-Based Bangla Sign Language Recognition with a Smart Glove Ongoing

*M.Sc. Thesis*, Institute of Appropriate Technology, BUET

- Developing a wearable sensor glove and IoT pipeline for BdSL recognition, supervised by Tahsina Farah Sanam, Ph.D.
- Integrating LLMs for sign-to-text and speech output; proposal approved by CASR
- Comparing flex-sensor and accelerometer/gyroscope (MPU-6050) sensing setups for the glove
- Building embedded ML gesture recognition for multimodal communication for deaf and mute individuals

### SMMNet-FedHE: Privacy-Preserving Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Accepted, IEEE TENCON 2026

*Postgraduate Research*, with A. Islam and T. F. Sanam

- Combined federated learning and homomorphic encryption for privacy-preserving, contactless detection of stereotypical motor movements for early autism screening
- Extracted multi-domain features from WiFi CSI data; accuracy above 80% across centralized and federated settings

### Wi-SMM: Stereotypical Motor Movement Detection for Autism Screening Using WiFi CSI Submitted

*Postgraduate Research*, with M. Mohtasim, A. Islam, T. F. Sanam

- Built a WiFi CSI-based dataset and detection framework for early, contactless ASD screening

### Wi-Fi CSI-Based Human Activity Detection Ongoing

*Independent Project*

- Capturing Wi-Fi channel state information (CSI) with two ESP32 boards for activity classification
- Developing the capture firmware with ESP-IDF on macOS

### ScabieSense: Scabies Detection, Treatment Guidance, and Outbreak Mapping Ongoing

*IAT, BUET, in collaboration with the Bangladesh Academy of Dermatology* (Principal Investigator: Tahsina Farah Sanam, Ph.D.)

- AI-assisted one-stop digital solution for scabies, developed under a BAD–IAT Memorandum of Agreement; supported by the International League of Dermatology and BAD
- Now in use by doctors screening students in different madrasas, the primary target group
- Researched the end-to-end workflow for hostel and school screening, case tracking, and follow-up
- Designed the clinical-AI pipeline, including questionnaire scoring, image analysis, and fallback logic
- Focused on mobile-friendly user experience, secure role-based access, and smooth case management

### BD Pedestrian and Overtaking Action Recognition for Driver Assistance

*Undergraduate Research Project*, North South University

- Collected and annotated 10+ hours of video dataset for deep learning models
- Developed computer vision algorithms for real-time action recognition
- Improved recognition accuracy for driver assistance systems in Bangladesh

### Hakuna Matata – Mental Health Social Network Platform

*Undergraduate junior Research Project*, North South University

- Designed user-centered interfaces using HTML, CSS, and JavaScript
- Conducted user research on mental health community engagement
- Applied HCI principles to create accessible digital environment

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## PUBLICATIONS AND MANUSCRIPTS

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- SMMNet-FedHE: Privacy-Preserving Deep Learning-Based Framework for Stereotypical Motor Movement Monitoring with WiFi CSI Data. A. Islam, A. S. M. S. Islam, T. F. Sanam. *Accepted, IEEE TENCON 2026*.
- Wi-SMM: Detection of Stereotypical Motor Movements for Early Screening of Autism Spectrum Disorder Using a WiFi CSI-Based Dataset. M. Mohtasim, A. Islam, A. S. M. S. Islam, T. F. Sanam. *Submitted*.

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## SERVICE

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## University Service

**Vice President**, NSU HR Club, North South University March 2022 – October 2023

- Organized “HR SPEAKS OUT 4.0” seminar with 700+ attendees
- Led IT, promotional, and strategic planning activities
- Managed recruitment of 300+ members

*Sub-Executive* (November 2020 – March 2022) | *Senior Member* (March 2020 – November 2020)

## Community Service

**Panel Deputy Speaker**, Child Parliament of NCTF, Bangladesh May 2016 – June 2018

- Prepared speakers for presentations to government ministers
- Facilitated pitches to Education Minister and Deputy Speaker of National Parliament

**Trainer and UDRT Member**, Bangladesh Red Crescent Society, Bogra Unit March 2015 – August 2016

- Completed disaster management training by Canadian Red Cross
- Led disaster relief operations in Sariakandi, Bogura

**Unit Leader**, Bogra Zilla School Red Crescent Unit October 2013 – October 2015

- Led team of 100+ active and 300 regular members
- Coordinated relief aid to 100+ flood-affected families

## LANGUAGES AND SKILLS

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### Languages

**Bengali:** Native | **English:** Professional Proficiency

### Technical Skills

**Programming:** Python, C/C++, Java, JavaScript, Swift

**Web Technologies:** HTML, CSS, Bootstrap, SQL

**Mobile Development:** iOS (Swift, Xcode)

**Design:** UI/UX Design, AutoCAD (2D/3D)

**Embedded Systems:** ESP32 (ESP-IDF), Arduino Uno, NodeMCU, MPU-6050, TCA9548A, Wi-Fi CSI

**Data Science:** Google Colab, NumPy, Pandas

**Tools:** Git, MS Office Suite, Google Workspace, LaTeX

## PROFESSIONAL DEVELOPMENT

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**Supervised Machine Learning: Regression and Classification** August 2025

DeepLearning.AI & Stanford University, Coursera

Credential: [coursera.org/verify/6I4AZ2IH4ANS](https://coursera.org/verify/6I4AZ2IH4ANS)

**Google UX Design Professional Certificate** Google, Coursera

7-course specialization: UX foundations, wireframing, prototyping, Figma

Credential: [lnkd.in/dDCmw4MP](https://lnkd.in/dDCmw4MP)

**The Complete 2021 Web Development Bootcamp** Udemy, September 2021

Credential: [UC-31cfe125-f6cb-405c-9432-e008b56b7f0f](https://www.udacity.com/course/complete-2021-web-development-bootcamp)

## REFERENCES

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**Tahsina Farah Sanam, Ph.D.**

Professor

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